

Standard Operating Procedure

Request-to-Exit (REX) & Door Position Switch Installation

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Related Documents	SOP-LV-015, SOP-LV-014
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Owner	Apex – Operations
Approved By	_____

1. Purpose

This procedure establishes standard practices for selecting, placing, wiring, and testing Request-to-Exit (REX) devices and door position switches (DPS) on access-controlled doors.

2. Scope

Applies to all Apex technicians installing REX sensors (PIR motion-based, push-button, or integrated into panic/crash bar hardware) and door position switches on access-controlled openings.

3. Why REX Matters — Life-Safety Function

REQUIRED, NOT OPTIONAL:

A REX device is required on every electromagnetic lock (maglock) installation, without exception. A maglock without a functioning REX path is a life-safety and code violation — it is not a convenience feature to be value-engineered out.

On an electric strike installation, a REX device serves a different but still important purpose: it allows the access control system to distinguish legitimate egress (someone using the door normally from the inside, via a crash bar, lever, or motion sensor) from an unauthorized or forced entry, suppressing a false door-forced alarm during normal use.

4. REX Device Types

- PIR (passive infrared) motion sensor — detects a person approaching or moving toward the door from the secure side; no physical action required by the person exiting.
- Push-button — a manual momentary switch mounted near the door; simple and highly reliable, but requires a deliberate action.
- Integrated into panic/crash bar hardware — the REX signal is generated mechanically by the bar itself as part of its normal operation.
- Selection depends on the application: PIR is common for general egress paths; a push-button or integrated crash-bar REX may be preferred where deliberate, confirmed egress action is wanted (e.g., higher-security doors).

5. PIR REX Placement

- Typical mounting height range is 7–15 ft (2.1–4.6 m) above the floor — confirm the specific model's rated range; mounting outside the rated height range degrades detection coverage.
- Aim and coverage pattern should be angled to detect a person approaching or standing at the door from the secure side, without extending detection into adjacent traffic areas that don't actually intend to use this door — false triggers from unrelated foot traffic are a common nuisance-alarm source.
- Do not locate a REX sensor where it could be defeated by reaching through a gap (e.g., between double doors) to trigger it from the unsecured side — this is a documented attack vector against PIR REX placement and must be considered during placement, not just detection range.

- After mounting, walk-test from all normal approach angles and adjust aim/sensitivity as needed. Also verify that traffic not intending to exit through this door does not typically trigger the detector.

6. REX Wiring

- Confirm whether the specific REX device output is a dry contact (N.O. or N.C.) or a powered/supervised output — wire to the access control panel's REX input accordingly; do not assume.
- Where the REX device has a door-contact input terminal, wire the door position switch (DPS) to it per the manufacturer's instructions — this typically enables a relay drop shortly after door close and activates a door-propped-open sounder function where supported.
- Where the panel/controller supports a REX shunt function, confirm the REX signal is wired to suppress the door-forced alarm for the correct shunt duration — verify this in testing (Section 8), not just that the REX contact closes.
- All access control circuits should be installed as Class 2 wiring per the applicable electrical code, using the wire gauge and type specified by the device manufacturer.

7. Door Position Switch (DPS) Installation

- Mount the DPS (typically a magnetic contact switch) so the door and frame halves align within the manufacturer's rated gap tolerance when the door is closed — excessive gap causes unreliable state detection.
- Confirm DPS polarity/state (N.O. vs. N.C.) matches what the panel or REX device expects — an incorrectly configured DPS can report door state backwards, masking a propped-open condition instead of alarming on it.
- Route DPS wiring to avoid mechanical stress at the door/frame hinge side, where repeated door operation over years can fatigue and break conductors at the flex point — use a door loop or sufficient slack, not a tight direct run.

8. Testing

- Test REX activation from all placement angles established during walk-testing (Section 5) — confirm the lock releases and/or the door-forced alarm is correctly suppressed.
- For maglock installations: confirm REX activation reliably releases the lock every time, not intermittently — a REX that only sometimes works is a life-safety failure waiting to happen, not a minor nuisance.
- Test the door-propped-open condition: hold the door open past the configured time and confirm the expected alarm/sounder behavior triggers.
- Confirm the DPS correctly reports door open/closed state in the access control software, matching the door's actual physical state through several open/close cycles.
- Record REX device type, placement, and DPS configuration on the site door schedule/configuration sheet.

9. References

- Manufacturer installation instructions for the specific REX/DPS model in use (govern where more restrictive than this SOP)
- UL 294 — Access Control System Units
- SOP-LV-015 — Electric Strike & Maglock Installation
- SOP-LV-014 — Fire Alarm Interface & Life-Safety Interlock